

Ladhe's Quad Conjecture: Every Prime Quadruplet Is an Additive/Multiplicative/Exponential Combination of Its Predecessors*

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Abstract

Every prime quadruplet after 2, 3, 5, 7 is a sum of the ones before it. Prime quadruplets are the constellations $(p, p + 2, p + 6, p + 8)$ of four primes, the densest possible packing of four primes beyond $(5, 7, 11, 13)$. We order them into a chain $\mathcal{Q}_1 = (11, 13, 17, 19)$, $\mathcal{Q}_2 = (101, 103, 107, 109)$, $\mathcal{Q}_3 = (191, 193, 197, 199)$, $\dots$, seeded by the *royal quad* $\mathcal{Q}_0 = (2, 3, 5, 7)$, and observe that the first member of every quadruplet can be written as the largest member of the previous quadruplet plus a sum of distinct members of earlier quadruplets, for example $2081 = 1879 + 199 + 3$ and $35551421 = 35545429 + 5659 + 199 + 109 + 19 + (2 \times 3)$; the other three members then follow without any search, as $p + 2$, $p + (2 \times 3)$ and $(p + 6) + 2$. The representation we record is fixed by one rule: after the base, always the largest earlier member that still leaves a solvable remainder, again and again; a royal value covers only what no member can. The precise form of the observation is that the gap D between a first member and its base is a sum of at most 8 distinct earlier members, with exactly four exceptional gaps, 22, 82, 172 and 382, that need a small expression in the royal members 2, 3, 5, 7. Members of earlier quadruplets are only ever added; $+$ and $\times$ act on nothing but the royal members, each used at most once and never more than two of them in a product. A congruence lemma modulo 210 shows that these four are the only exceptions in the entire admissible range below 10^9 , not merely the ones we met. Allowing one such expression in every gap brings the bound down to six terms with no exception at all. We call this *Ladhe's Quad Conjecture* and verify it for all 28,387 prime quadruplets below 10^9 with an independent checker. Two further representations of each gap, one that prefers multiplication and one that prefers powers of chain members, are also computed and released. We test the objection that the pattern is a consequence of density rather than of primality by running the same solver on control chains of identical spacing whose members are not prime: the bound of eight and the parity of the term counts are reproduced by a control that preserves the residue class, and are therefore structural, but no control reproduces the small closed set of exceptions. We also prove, from a two-line covering lemma and computed base cases, that the chain is an additive basis for all integers: every integer up to the next quadruplet (indeed up to a bound that grows like the sum of all earlier members) is a sum of distinct chain members plus one expression in 2, 3, 5, 7 in which, unlike in the conjecture itself, $-$ is allowed alongside $+$ and $\times$; this is unconditional below 10^9 and rests beyond that only on a mild growth condition. We explain carefully what the conjecture does and does not say about locating the next quadruplet, give a heuristic for why such representations

*Correspondence: spalgorithm@gmail.com. Paper and dataset are archived on Zenodo, <https://doi.org/10.5281/zenodo.22286846> (this DOI always resolves to the latest version), and released under CC BY 4.0; the code is released under the licence of the repository <https://github.com/SPAlgorithm/LE>.

are plentiful, and state open questions. The utility that builds the chain and the full dataset are available at <https://github.com/SPAlgorithm/LE>.

1 Introduction

A *prime quadruplet* is a quadruple of primes of the form

$$(p, p + 2, p + 6, p + 8),$$

the tightest admissible pattern of four primes. Apart from $(5, 7, 11, 13)$, every prime quadruplet has $p \equiv 11 \pmod{30}$, so its members end in the digits 1, 3, 7, 9 in that order. The sequence of first members 11, 101, 191, 821, 1481, 1871, 2081, ... is OEIS A007530 [3]. Whether there are infinitely many prime quadruplets is open; it is the case $k = 4$ of the Hardy–Littlewood prime k -tuple conjecture [1], and the Bateman–Horn heuristic [2] predicts that the number of quadruplets up to x is asymptotically $C_4 x / (\log x)^4$ with $C_4 \approx 4.15$. The strongest unconditional results in this direction are the bounded-gap theorems of Zhang [4] and Maynard [5], which give infinitely many pairs (and m -tuples) of primes within a bounded distance but not the specific pattern $(0, 2, 6, 8)$. Large quadruplets and related clusters have been searched for computationally for decades [6].

This note records an additive structure that runs through the whole sequence of prime quadruplets, at least as far as we have looked. Order the quadruplets by size and call the k -th one $\mathcal{Q}_k$; put the four primes 2, 3, 5, 7 in front as $\mathcal{Q}_0$, the *royal quad*. Then the first member of $\mathcal{Q}_k$ is obtained from the largest member of $\mathcal{Q}_{k-1}$ by adding a few distinct members of earlier quadruplets, and the other three members of $\mathcal{Q}_k$ are that first member plus 2, plus 2×3 , and plus $2 \times 3 + 2$. In one line:

Every prime quadruplet after 2, 3, 5, 7 is a sum of the ones before it.

Some representation by a plain sum of earlier members, with at most one added royal value, exists for every first member below 10^9 except 11 and the 30 first members whose gap is 82, which need one product of royal members, as in $101 = 19 + 17 + 13 + 11 + (5 \times 7) + (2 \times 3)$ (Section 6). The representation the utility stores, chosen by the largest-first rule of Definition 5, is a plain sum for 18,131 of the 28,387 first members and closes with one such product for the rest. Section 3 states this precisely in three forms of increasing strength, Section 4 proves a companion statement, that with subtraction allowed among the royal members every integer below the next quadruplet is such a sum, Section 5 walks through examples that can be checked by hand, Section 6 reports the computation up to 10^9 , and Section 8 discusses, with some care, what the statement means for finding the next quadruplet.

The observation grew out of the authors' earlier work on additive decompositions of primes, *Ladhe's Conjecture* [7], which concerns single primes written as sums of three primes and underlies a hash-based signature scheme [8]. The present note is about prime *quadruplets* and is self-contained.

2 Definitions and rules

Definition 1 (Quads). $\mathcal{Q}_0 = (2, 3, 5, 7)$ is the royal quad. For $k \geq 1$, $\mathcal{Q}_k$ is the k -th prime quadruplet $(p, p + 2, p + 6, p + 8)$ with $p \geq 11$, in increasing order of p . We write its members as $q_k^{(1)} = p$, $q_k^{(3)} = p + 2$, $q_k^{(7)} = p + 6$, $q_k^{(9)} = p + 8$, the superscript being the last decimal digit. The quadruplet $(5, 7, 11, 13)$ is not part of the chain, because 5 and 7 already belong to $\mathcal{Q}_0$, and $\mathcal{Q}_0$ itself is not of the form $(p, p + 2, p + 6, p + 8)$.

Definition 2 (Base, gap, pool). For $k \geq 2$ the base of $\mathcal{Q}_k$ is $m_k = q_{k-1}^{(9)}$, the largest member of the previous quadruplet. For a member t of $\mathcal{Q}_k$ the gap is $D = t - m_k > 0$. The pool available to $\mathcal{Q}_k$ is $\mathcal{P}_k = \mathcal{Q}_1 \cup \dots \cup \mathcal{Q}_{k-1}$, the set of all $4(k-1)$ primes of the earlier quadruplets.

Definition 3 (Royal expressions). A royal expression is an arithmetic expression built from the members 2, 3, 5, 7 of $\mathcal{Q}_0$, each used at most once, with the operations + and $\times$ only, where a product may multiply only two members: $(5 \times 7) + (2 \times 3)$ is a royal expression, $2 \times 5 \times 7$ and $5 \times (7 + (2 \times 3))$ are not. Let $\mathcal{R}$ be the set of its values; $\mathcal{R}$ has 30 elements between 2 and 41, and every one of them is flat (no product contains a sum). One subset matters below: $\mathcal{R}_+$, the 11 values reachable by addition alone, $\{2, 3, 5, 7, 8, 9, 10, 12, 14, 15, 17\}$.

Definition 4 (Additive representation). An additive representation of a member t of $\mathcal{Q}_k$ ($k \geq 2$) is an identity

$$t = m_k + s_1 + s_2 + \cdots + s_j + r, \quad s_1 > s_2 > \cdots > s_j, \quad s_i \in \mathcal{P}_k, \quad r \in \mathcal{R} \cup \{0\},$$

with all s_i distinct and none equal to m_k . Members of earlier quadruplets are thus only ever added; multiplication occurs only inside the royal expression r , and only between two of 2, 3, 5, 7. Its length is j if $r = 0$ and $j + 1$ otherwise. Several s_i may come from the same quadruplet. (The stricter reading “at most one member per quadruplet” already fails at $\mathcal{Q}_2$: $101 - 19 = 82$, and no single member of $\mathcal{Q}_1$ leaves a remainder in $\mathcal{R}$.) A member t of $\mathcal{Q}_1$ is written as a royal expression alone, e.g. $11 = (2 \times 3) + 5$.

Only the first member p of $\mathcal{Q}_k$ is given such a representation by search. Its three companions are written

$$p + 2 = p + 2, \quad p + 6 = p + (2 \times 3), \quad p + 8 = (p + 6) + 2,$$

which is all the quadruplet pattern requires of them; they are neither searched for nor listed. From here on “the gap” of $\mathcal{Q}_k$ means the gap $D = p_k - m_k$ of its first member.

Definition 5 (Canonical representation). Among the additive representations of a first member the utility stores the one given by the following rule. Start with $r = D$. While $r > 0$ and some unused member of $\mathcal{P}_k$ does not exceed r , take the largest such member and continue with the remainder; when a choice leads to a remainder that cannot be finished, the next smaller member is tried instead (the search backs up). When no unused member fits, the remainder must be 0 or a value in $\mathcal{R}$, which closes the equation. The terms $s_1 > s_2 > \cdots$ are thus chosen greedily, largest first, and a royal value covers only what no member can. For the shared equation of a gap the base is among the candidates (any quadruplet reusing the gap has its base above it); a first member whose base lies below its own gap, which happens for 101 and 821 only, keeps an equation of its own with the base excluded. The bounds of Section 3 concern the shortest representations, which a separate search computes (Section 6); the canonical form is usually longer.

The following congruences are elementary but shape everything below.

Lemma 1. Let $\mathcal{Q}_k = (p, p + 2, p + 6, p + 8)$ be a prime quadruplet with $p \geq 11$ and write $M = p + 4$, so that its members are $M - 4, M - 2, M + 2, M + 4$. Then

- (i) $M \equiv 15 \pmod{30}$, equivalently $p \equiv 11 \pmod{30}$;
- (ii) $M \not\equiv \pm 2, \pm 4 \pmod{7}$, hence $M \equiv 15, 105, 195 \pmod{210}$ and

$$p \equiv 11, 101, 191 \pmod{210},$$

equivalently $p \equiv 2, 3, 4 \pmod{7}$;

- (iii) consecutive first members differ by $h \equiv 0, 30, 90, 120, 180 \pmod{210}$;
- (iv) the gaps of the four members of $\mathcal{Q}_k$ satisfy

$$D \equiv 22, 24, 28, 0 \pmod{30} \quad \text{for } t = q_k^{(1)}, q_k^{(3)}, q_k^{(7)}, q_k^{(9)} \text{ respectively,}$$

and, more precisely, D lies in one of the twenty classes

$$\begin{aligned} D \equiv & 0, 22, 24, 28, 30, 82, 84, 88, 90, 112, \\ & 114, 118, 120, 172, 174, 178, 180, 202, 204, 208 \pmod{210}. \end{aligned}$$

Since only the first member is searched for, the gaps that actually occur are those of $t = q_k^{(1)}: D \equiv 22 \pmod{30}$, in one of the five classes 22, 82, 112, 172, 202 $\pmod{210}$, the five classes 22, 82, 112, 172, 202 being those of a first member;

- (v) every gap is even while every pool member is odd, so a representation with $r = 0$ has an even number of terms.

Proof. All four members are primes larger than 7, hence coprime to 2, 3, 5, 7. Modulo 3 they are $M + 2, M + 1, M + 2, M + 1$, so $M \equiv 0$; modulo 5 they are $M + 1, M + 3, M + 2, M + 4$, so $M \equiv 0$; and M is odd. This is (i). Modulo 7 none of $M \pm 2, M \pm 4$ may vanish, so $M \equiv 0, \pm 1$, and combining with (i) leaves the three classes of (ii). Part (iii) is the set of differences of $\{11, 101, 191\}$ modulo 210. For (iv), writing $\delta \in \{0, 2, 6, 8\}$ for the offset of t inside $\mathcal{Q}_k$,

$$D = (p_k + \delta) - (p_{k-1} + 8) \equiv h + \delta - 8 \pmod{210},$$

and the four values of δ give the four classes modulo 30, the five values of h from (iii) the twenty classes modulo 210. Finally D is a difference of two odd numbers, hence even, and a sum of j odd numbers is congruent to j modulo 2, which is (v). $\square$

Pool members are $\equiv 11, 13, 17, 19 \pmod{30}$, hence a two-term representation of the gap of a first member ($D \equiv 22$) must add two first members ($11 + 11 \equiv 22$).

Three ways. The utility described in Section 6 computes, for every member, one representation of each of three kinds:

- A** *additive*, as in the definition above, chosen so that addition is exhausted before any royal multiplication is used (fewer multiplications first, then fewer terms, then larger members first);
- M** *multiplicative*: any two distinct members of $\mathcal{P}_k \cup \mathcal{Q}_0$ may be multiplied, and multiplication is preferred over addition (the largest unused member is multiplied by the largest member that still fits, then the next, plain addition only when nothing can be multiplied);
- E** *exponential*: in addition, a member may be raised to a member power, powers being preferred over products over plain addition. Because the order in which powers are tried changes the outcome, two variants are kept: E1 takes the largest base first, E2 the largest power value first;
- M1** *without the base*: the M search applied to the member itself instead of to its gap. No base term is required, the pool is every member of $\mathcal{P}_k \cup \mathcal{Q}_0$ below the target, and the largest products carry the number, e.g. $854921 = (427249 \cdot 2) + (109 \cdot 3) + (17 \cdot 5) + 11$;
- E3** *powers first*: the largest power b^e below the member, with b and e distinct members, then again the largest power below what is left as long as it covers at least half of it, followed by the remainder in the additive way (members below the target added, largest first, closed by a flat royal expression), e.g. $854921 = 829^2 + 11^5 + 17^3 + 1489 + 107 + 101 + 19$ and $19497221 = 11^7 + 2^{13} + 829 + 827 + 199 + 3$. If no choice of members closes the remainder, the chain of powers is shortened by one, and finally the next largest first power is tried.

In A, M, E1 and E2 every member of $\mathcal{P}_k \cup \mathcal{Q}_0$ is used at most once and the base m_k is always a plain term; M1 and E3 also use every member at most once, the base and exponent of the E3 power included, but have no base term.

3 The conjecture

We state the observation in three forms. The first is the one the utility works with; the second and third are the ones we consider interesting, because they bound the number of terms.

Conjecture 1 (Ladhe’s Quad Conjecture, existence form). *The first member p_k of every prime quadruplet $\mathcal{Q}_k$ with $k \geq 2$ has an additive representation over the chain: $p_k = m_k + \sum_i s_i + r$ with distinct $s_i \in \mathcal{P}_k$ and $r \in \mathcal{R} \cup \{0\}$; the other three members then follow as $p_k + 2$, $p_k + (2 \times 3)$ and $(p_k + 6) + 2$.*

Conjecture 2 (bounded pure form). *There is an absolute constant K such that, apart from finitely many gaps, every gap $D = p_k - m_k$ ($k \geq 2$) is a sum of at most K distinct members of $\mathcal{P}_k$. The data support $K = 8$, with exactly four exceptional gaps:*

$$D \in \{22, 82, 172, 382\},$$

each of which needs a royal value. No new exceptional gap occurs beyond $\mathcal{Q}_{435}$ (first member 3,512,231) up to 10^9 , and the list is in fact complete in a stronger sense than observation: by Lemma 1 every gap lies in one of five classes modulo 210, and among all 9,523 even numbers in those classes up to 400,000—a range that covers every gap below 10^9 , the largest being 395,512—exactly these four are not sums of at most eight distinct chain members below themselves. Four further small values, 52, 262, 472 and 772, are equally unreachable but can never occur as gaps, their classes being excluded by the lemma. The last occurrence of an exceptional gap below 10^9 is at $\mathcal{Q}_{28063}$ (first member 985,124,051). In all, 98 of the 28,386 gaps are exceptional.

Conjecture 3 (bounded form with one flat royal value). *Every gap $D = p_k - m_k$ ($k \geq 2$) is a sum of at most five distinct members of $\mathcal{P}_k$ plus at most one element of $\mathcal{R}$; that is, every first member has an additive representation of length at most six.*

Remark 1 (Why the existence form alone is weak). *Once a few quadruplets exist, the pool is large and the sums of its subsets cover every even number in range many times over, so Conjecture 1 is very likely to hold for reasons that have little to do with quadruplets; it would also be vacuous if only finitely many quadruplets existed. What is not automatic is that a fixed number of terms suffices while the gaps grow: the average gap between consecutive quadruplets near x grows like $(\log x)^4/C_4$, about 35,000 at 10^9 , and the pool members that fit under a gap are few (92 below the average gap and 336 below the largest gap, 395,520, in our data). Conjectures 2 and 3 would be falsified by a single gap that is not a short sum; the natural falsification test is a very large gap between consecutive quadruplets.*

Remark 2 (The exceptional gaps). *All four exceptions are small and appear early. The gap 82 belongs to $\mathcal{Q}_2$ (and reappears whenever two consecutive quadruplets are 90 apart, as at $\mathcal{Q}_3$, $\mathcal{Q}_{14}$, $\mathcal{Q}_{437}$, ...); at that point the pool is $\mathcal{Q}_1$ alone, whose subset sums cannot reach 82. The gaps 22, 172 and 382 are coverage holes of $\mathcal{Q}_1 \cup \mathcal{Q}_2$: for instance $382 = 1871 - 1489$ is reached by no subset of the members below it; its stored form $199 + 109 + 19 + 17 + 13 + 11 + 7 + 5 + 2$ closes with the royal sum 14. With the 11 royal addition values allowed, only 82 remains exceptional; with all of $\mathcal{R}$ allowed there is no exception. In the light of Lemma 1 the four are exactly the coverage holes that survive the congruence filter: of the eight values below 400,000 that no sum of at most eight distinct members reaches, four lie in classes no gap can occupy. Within the admissible classes the pressure relaxes quickly: below 400,000 six terms suffice for every value above 1552, and four for every value above 53,362. Appendix B lists the four gaps with their forced representations.*

4 Coverage: every integer below the next quadruplet

The chain also serves as an additive basis for *all* integers, not only for the quadruplets, once subtraction, and products of any shape, are allowed inside the royal part. The two settings should not be confused: the conjecture writes each quadruplet with $+$ and $\times$ only, at most two royal members to a product; the derivation of an arbitrary integer below also uses $-$, and its royal part may be nested. This section proves the second statement from two computed base cases.

Definition 6 (Signed royal values). $\mathcal{R}^\pm$ is the set of non-negative values of expressions built from the members 2, 3, 5, 7 of $\mathcal{Q}_0$, each used at most once, with $+$, $-$ and $\times$ placed anywhere; unlike $\mathcal{R}$, there is no restriction to two-factor products. It has 83 elements between 0 and 210, contains every integer from 0 to 38, and its first gaps are 39, 43, 53, 59, 61, 62. Examples: $0 = 5 - (3 + 2)$, $1 = 3 - 2$, $4 = 7 - 3$, $33 = (5 \times 7) - 2$, $67 = (2 \times 5 \times 7) - 3$. The utility reaches for $\mathcal{R}$ first and for the signed values only when no choice of primes allows a closing in $\mathcal{R}$ ($928 = 823 + 103 + 2$ rather than $829 + 19 + 13 + (2 \times 5 \times 7) - 3$); the values with three-factor or nested products are the last resort of all. Below 300,000 that resort is taken for exactly one integer: $99 = 19 + 13 + 67$ has no derivation with two-factor products, since every remainder $99 - s$ over subsets s of $\{11, 13, 17, 19\}$ misses the two-factor values. A nested product (a sum inside a product, such as $5 \times ((2 \times 7) - 3)$) is never used.

Definition 7 (Derivation). A derivation of an integer $n \geq 0$ over $\mathcal{Q}_0, \dots, \mathcal{Q}_k$ is an identity $n = s_1 + \dots + s_j + r$ with distinct $s_i \in \mathcal{Q}_1 \cup \dots \cup \mathcal{Q}_k$ and $r \in \mathcal{R}^\pm$ ($j = 0$ or $r = 0$ allowed). Quadruplet members are only added; subtraction and multiplication occur only among royal members. Write C_k for the set of integers that have a derivation over $\mathcal{Q}_0, \dots, \mathcal{Q}_k$ and T_k for the largest T with $[0, T] \subseteq C_k$.

By direct computation,

$$T_0 = 38, \quad T_1 = 156, \quad T_2 = 576, \quad T_3 = 1356, \quad T_4 = 4656,$$

so with the pool 2, 3, 5, 7, 11, 13, 17, 19 every integer from 0 to 156 has a derivation, in particular every integer up to 101; for instance $16 = 11 + 5$, $53 = 19 + 11 + (7 \times 5) - 2$ and $55 = 19 + 11 + (7 \times 5)$.

Lemma 2. Let A be a set of non-negative integers containing every integer in $[0, T]$, and let $0 < m \leq T + 1$. Then $A \cup (A + m)$ contains every integer in $[0, T + m]$.

Proof. An integer $x \in [0, T]$ lies in A . An integer $x \in [T + 1, T + m]$ satisfies $0 \leq x - m \leq T$, so $x - m \in A$ and $x \in A + m$. $\square$

Proposition 1. Put $S_3 = T_3 = 1356$ and $S_k = S_{k-1} + 4p_k + 16$ for $k \geq 4$. If $p_j \leq S_{j-1} + 1$ for every j with $4 \leq j \leq k + 1$, then $T_k \geq S_k \geq p_{k+1}$: every integer n with $0 \leq n \leq p_{k+1}$ has a derivation over $\mathcal{Q}_0, \dots, \mathcal{Q}_k$. For $k \leq 3$ the same conclusion holds by the computed values ($T_0 \geq p_1 = 11$, $T_1 \geq p_2 = 101$, $T_2 \geq p_3 = 191$, $T_3 \geq p_4 = 821$).

Proof. Induction on $k \geq 3$ with the claim $T_k \geq S_k$; the case $k = 3$ is the computation. Assume $T_{k-1} \geq S_{k-1}$ and add the members $p_k, p_k + 2, p_k + 6, p_k + 8$ of $\mathcal{Q}_k$ one at a time, each with coefficient 0 or 1. The first satisfies $p_k \leq S_{k-1} + 1 \leq T_{k-1} + 1$ by hypothesis, so Lemma 2 raises the coverage to $T_{k-1} + p_k$; each following member is at most $p_k + 8$, far below the coverage reached so far, so the lemma applies three more times and $T_k \geq T_{k-1} + 4p_k + 16 \geq S_k$. Finally $p_{k+1} \leq S_k + 1$ and p_{k+1} is odd while S_k is even, hence $p_{k+1} \leq S_k$. $\square$

The hypothesis is mild: it fails only if a quadruplet's first member exceeds $S_{j-1} + 1$, which grows like four times the sum of all earlier first members. In the chain up to 10^9 it holds at every $j \leq 28,387$, and it is tightest at $j = 4$ ($p_4 = 821$ against $S_3 + 1 = 1357$). The one-step condition

$p_j \leq 5p_{j-1} + 16$ is sufficient and also holds throughout (the largest ratio p_j/p_{j-1} for $j \geq 3$ is 4.30, at $j = 4$). So Proposition 1 is unconditional below 10^9 ; an unconditional proof for all k would follow from any Bertrand-type bound for prime quadruplets, which is open, although far more than what is needed here.

Remark 3. *Coverage is a statement about all integers with subtraction allowed and does not imply Conjecture 1, which forbids subtraction, fixes the base m_k and, in its bounded forms, limits the number of terms. It does show that the differences D of the conjecture, which are tiny compared with T_{k-1} , always have some derivation of this looser kind.*

The utility prints a derivation of any integer with `-d`, taking the largest quad prime below the remainder first, each prime once, and closing in the vocabulary of the chain equations whenever the choice of primes allows it: royal members added if that suffices, else one product of two members, else two, the primes being re-chosen between those passes. Only when no choice of primes closes with $+$ and $\times$ does a minus enter, and only when even that fails a product of three. A range `-d 53 83` prints one line per integer:

```
python3 lc.py -d 16      16 = 13 + 3
python3 lc.py -d 53      53 = 19 + 17 + 7 + 5 + 3 + 2
python3 lc.py -d 55      55 = 19 + 17 + 11 + 5 + 3
python3 lc.py -d 76      76 = 19 + 17 + (5*7) + 3 + 2
python3 lc.py -d 99      99 = 19 + 13 + (2*5*7) - 3
python3 lc.py -d 101     101 = 19 + 17 + 13 + 11 + (5*7) + (2*3)
python3 lc.py -d 999986171
                        999986171 = 999900529 + 82729 + 2089 + 821 + 3
```

The greedy choice is not the only derivation; $53 = 19 + 17 + 13 + 7 - 3$ and $55 = 19 + 11 + (7 \times 5)$ are equally valid. Below 300,000 exactly nine integers need a minus at all — 0, 1, 4, 92, 93, 94, 96, 99 and 102, each with only $\mathcal{Q}_1$ or $\mathcal{Q}_1 \cup \{101\}$ to draw on — and one, 99, a product of three royal members.

5 Worked examples

The lines below are copied verbatim from the utility (`python3 lc.py 12 -AME`); A, M, M1, E1, E2, E3 label the ways, and a line is omitted when it repeats an earlier line of the same member (E2 when it coincides with E1; in $\mathcal{Q}_1$, which has no base, M1 coincides with M). Products are written in parentheses and $b \wedge e$ means b^e .

$\mathcal{Q}_1 = (11, 13, 17, 19)$ **from the royal quad.** The first quadruplet is written with royal expressions only; the utility also lists alternatives, including ones with subtraction, which the chain equations of Section 3 otherwise never use.

```
11 = (2*3) + 5      also: (2*7) - 3 | (3*7) - (2*5) | 7 + 5 + 2 - 3
13 = (2*5) + 3
17 = 7 + 5 + 3 + 2
19 = (2*7) + 5
```

$\mathcal{Q}_2 = (101, 103, 107, 109)$: **the gap 82.** Base $m_2 = 19$, gap $101 - 19 = 82$. The pool is $\mathcal{Q}_1$, whose members other than the base sum to 41, so a royal value is unavoidable; 41 itself is flat, $(5 \times 7) + (2 \times 3)$.

```
A: 101 = 19 + 17 + 13 + 11 + (5*7) + (2*3)
M: 101 = 19 + (17*3) + (13*2) + 5
M1: 101 = (19*5) + (3*2)
E1: 101 = 19 + (7^2) + (11*3)
E3: 101 = (7^2) + 19 + 17 + 13 + 3
```

$Q_3 = (191, 193, 197, 199)$: **the same gap with a larger pool.** $191 - 109 = 82$ is the gap of Q_2 again, but the base is now 109 and 19 is free, so the rule takes it first: $82 = 19 + 17 + 13 + 11 + 22$ with the flat royal $22 = (3 \times 5) + 7$, one product where Q_2 needed two. The utility stores one equation per distinct gap, searched over every member below the gap; a member whose base lies *below* its own gap, as at Q_2 and Q_4 , cannot use that base as a term and keeps its own equation instead. Below 10^9 only two first members do, 101 and 821.

$$\begin{aligned} \text{A: } 191 &= 109 + 19 + 17 + 13 + 11 + (3 \times 5) + 7 \\ \text{M: } 191 &= 109 + (19 \times 3) + (7 \times 2) + 11 \\ \text{M1: } 191 &= (19 \times 7) + (17 \times 3) + 5 + 2 \\ \text{E1: } 191 &= 109 + (7 \times 2) + (11 \times 3) \\ \text{E3: } 191 &= (13 \times 2) + (5 \times 3) + 7 \end{aligned}$$

$Q_4 = (821, 823, 827, 829)$: **the largest members first.** Gap $821 - 199 = 622 = 197 + 193 + 191 + 19 + 17 + 5$; the three remaining members of Q_3 are taken in turn, then 19 and 17 from Q_1 , and the remainder 5, which no unused member fits, is royal. (The royal value $41 = (5 \times 7) + (2 \times 3)$ would have closed the gap two terms earlier; the rule keeps taking members while any fits.) The two exponential variants differ here: E1 starts from the largest base 19^2 , E2 also, but then prefers $3^5 = 243$ over $5^3 = 125$.

$$\begin{aligned} \text{A: } 821 &= 199 + 197 + 193 + 191 + 19 + 17 + 5 \\ \text{M: } 821 &= 199 + (197 \times 3) + (13 \times 2) + 5 \\ \text{M1: } 821 &= (199 \times 3) + (103 \times 2) + 13 + 5 \\ \text{E1: } 821 &= 199 + (19^2) + (5^3) + 101 + 17 + 11 + 7 \\ \text{E2: } 821 &= 199 + (19^2) + (3^5) + 11 + 7 \\ \text{E3: } 821 &= (19^2) + (7^3) + 101 + 11 + 5 \end{aligned}$$

$Q_6 = (1871, \dots)$: **a long gap early in the chain.** Gap $1871 - 1489 = 382$ is one of the four exceptions: no subset of $Q_1 \cup Q_2$ sums to 382, so a royal value is needed. The shortest representation with a royal *addition* value has seven terms ($199 + 109 + 19 + 17 + 13 + 11 + 14$, with $14 = 7 + 5 + 2$); with one royal product five terms suffice, $382 = 199 + 109 + 19 + 17 + 38$ with $38 = (5 \times 7) + 3$. The largest-first rule gives exactly this seven-term form. The multiplicative way is one product.

$$\begin{aligned} \text{A: } 1871 &= 1489 + 199 + 109 + 19 + 17 + 13 + 11 + 7 + 5 + 2 \\ \text{M: } 1871 &= 1489 + (191 \times 2) \\ \text{M1: } 1871 &= (829 \times 2) + (19 \times 7) + (17 \times 3) + 13 + 11 + 5 \\ \text{E1: } 1871 &= 1489 + (19^2) + (7 \times 3) \\ \text{E3: } 1871 &= (11^3) + 199 + 197 + 109 + 17 + 13 + 5 \end{aligned}$$

$Q_{12} = (13001, \dots)$: **a two-term gap, and what the rule makes of it.** Gap $13001 - 9439 = 3562 = 3461 + 101$, one member of Q_9 and one of Q_2 , both first members as the congruence $22 \equiv 11 + 11 \pmod{30}$ requires. About one gap in four has such a two-term representation. The stored equation is longer: the rule takes 3467, the largest member below 3562, and the remainder 95 then needs $19 + 17 + 13 + 11$ and (5×7) .

$$\begin{aligned} \text{A: } 13001 &= 9439 + 3467 + 19 + 17 + 13 + 11 + (5 \times 7) \\ \text{M: } 13001 &= 9439 + (1489 \times 2) + (193 \times 3) + 5 \\ \text{M1: } 13001 &= (5659 \times 2) + (199 \times 7) + (19 \times 13) + (5 \times 3) + 17 + 11 \\ \text{E1: } 13001 &= 9439 + (19^2) + (13^3) + (109 \times 7) + 107 + 101 + 17 + 11 + 5 \\ \text{E2: } 13001 &= 9439 + (13^3) + (19^2) + (109 \times 7) + 107 + 101 + 17 + 11 + 5 \\ \text{E3: } 13001 &= (109^2) + 829 + 193 + 19 + 17 + 13 + 11 + (7 \times 5) + 3 \end{aligned}$$

Two consecutive quadruplets near 3.5×10^7 . Q_{2208} and Q_{2209} are 6,000 apart; the gap of the first, $35545421 - 35524459 = 20962$, is $19429 + 1489 + 19 + 17 + 5 + 3$, and the gap of the second, $35551421 - 35545429 = 5992$, is $5659 + 199 + 109 + 19 + (2 \times 3)$.

A: $35545421 = 35524459 + 19429 + 1489 + 19 + 17 + 5 + 3$
 M: $35545421 = 35524459 + (9439*2) + (199*7) + (193*3) + (19*5) + 17$
 M1: $35545421 = (17771269*2) + (829*3) + (19*13) + (7*5) + 103 + 11$
 E1: $35545421 = 35524459 + (109^2) + (19^3) + (199*7) + (107*5) + 193 + 101$
 E3: $35545421 = (5659^2) + (19^5) + (101^3) + 13009 + 1489 + 109 + 103 + 17 + 13$

A: $35551421 = 35545429 + 5659 + 199 + 109 + 19 + (2*3)$
 M: $35551421 = 35545429 + (2089*2) + (199*7) + (103*3) + (19*5) + 17$
 M1: $35551421 = (17773369*2) + (1489*3) + (19*11) + 7$
 E1: $35551421 = 35545429 + (19^2) + (17^3) + (103*5) + (13*7) + 101 + 11$
 E3: $35551421 = (5659^2) + (19^5) + (101^3) + 19423 + 823 + 193 + 191 + 103 + 7$

The last quadruplet below 10^9 is $\mathcal{Q}_{28387} = (999986171, \dots)$ with $999986171 = 999900529 + 82729 + 2089 + 821 + 3$, and the largest gap between consecutive quadruplets in that range, 395,520 at $\mathcal{Q}_{19558}$, still has a two-term representation, $395512 = 392261 + 3251$; the stored equation is $628641701 = 628246189 + 394819 + 199 + 197 + 193 + 101 + 3$.

6 Computational evidence

The utility. `lc.py` is a single Python 3 file with no dependencies. It finds quadruplets with a segmented sieve (only $p \equiv 11 \pmod{30}$ is tested), then writes every first member's gap by the largest-first rule of Definition 5, backing up when a choice leads nowhere (the minimal lengths reported below come from a separate iterative-deepening search), and stores one set of A/M/E1/E2 equations per distinct gap, plus the base-free M1 and E3 equations of every first member, in a JSON cache that later runs `reuse`; the other three members of a quadruplet are recorded in their fixed form only. The additive search is capped at 16 terms and 300,000 nodes per call, the M/E searches at 50,000 nodes, the base-free M1 search, whose pool is the whole chain, at 1,000,000 nodes, and the additive remainder of E3 at 300,000 nodes; a fallback path exists for the case that a search gives up, and it was never taken in the data reported here (zero fallback flags). Building the chain to 10^9 took 43 s on a 2019 Intel laptop under Python 3.13.5; the base-free M1 and E3 equations, computed once per first member over the whole chain, account for most of it.

The checker. `check.py` re-reads the JSON and, independently of the search code, verifies every stored equation: the arithmetic, that all members are distinct, that the base is never reused as a term, that every term is smaller than the target, that every royal expression uses each of 2, 3, 5, 7 at most once and multiplies at most two of them, and that the second, third and fourth member of every quadruplet are stored in their fixed form, and that every stored additive equation is the one the largest-first rule of Definition 5 produces, re-deriving it with a search of its own. On the 10^9 dataset it checked 170,340 equations (6 per first member, the four members of $\mathcal{Q}_1$ included), 3,421 royal expressions, 85,158 fixed forms and 3,411 re-derived additive equations, and found 0 errors.

Minimal lengths. `stats.py` recomputes, for every distinct gap and independently of the preferences of the utility, the minimal length K of a representation under three vocabularies: pool members only (*pure*); pool members plus at most one value of $\mathcal{R}_+$; pool members plus at most one value of $\mathcal{R}$. The minimum is taken when the gap is first met, using only the quadruplets available at that moment. Table 1 gives the result.

The pure column has only even lengths, as it must. The four gaps without a pure representation are those of Conjecture 2; with $\mathcal{R}_+$ only 82 remains; with $\mathcal{R}$ nothing remains and six terms always suffice, which is Conjecture 3. The mean minimal length does not grow along the chain: in eight consecutive blocks of about 3,550 quadruplets the mean pure K of the first

Table 1: Minimal length K of a representation of the gap, over the 3,409 distinct first-member gaps met below 10^9 .

vocabulary	$K=1$	2	3	4	5	6	7	8	none	mean
pure members	–	687	–	2608	–	104	–	6	4	3.66
+ one value of $\mathcal{R}_+$	–	688	429	2179	101	5	6	–	1	3.51
+ one value of $\mathcal{R}$	1	705	425	2249	27	2	–	–	0	3.47

member’s gap is 3.80, 3.75, 3.73, 3.71, 3.71, 3.67, 3.73, 3.71 and the share of two-term gaps stays between 22% and 24%.

What the utility actually stores. Table 2 summarises the additive representations chosen by the utility. Because the rule always takes the largest member that fits and leaves only what no member covers to the royal quad, these are usually longer than the minimum: 10,255 of the 28,386 first-member equations close with a royal product, 14,709 with an added royal value, and 3,422 are pure sums of pool members. The longest stored equation has twelve terms after the base, the mean 5.8. Of the 28,386 gaps, 24,977 repeat a gap seen earlier in the chain; there are only 3,409 distinct gaps, because consecutive quadruplets are often the same distance apart.

Table 2: Additive representations stored by the utility, 10^9 dataset. Length counts the terms after the base, a royal value counting as one.

scope	length											royal part
	2	4	5	6	7	8	9	10	11	12		none / + / ×
first members	683	6962	6045	4077	6479	1578	2211	159	180	12	3422 / 14709 / 10255	

Multiplicative and exponential ways. For the 28,386 first-member gaps, the M representations use 90,661 products and 50,439 plain terms, never more than nine terms; 3,005 of the products multiply two royal members. E1 uses 60,176 powers, 49,728 products and 79,246 plain terms (at most 14 terms); E2 uses 63,213 powers, 34,739 products and 78,861 plain terms (at most 12). E1 and E2 coincide for 9,117 of the 28,386 gaps. These representations are released for further study; we make no claim about them beyond their existence within the search caps. The base-free M1 and powers-first E3 representations were likewise found for every first member and for the four members of $\mathcal{Q}_1$ (only 11, 17 and 19, which are written from the royal quad alone, admit no power chain and fall back to the power-first search) and are stored and verified. Of these 28,390 E3 chains, 4,231 have one power, 8,950 two, 11,636 three and 3,570 four; we do not analyse these representations further here.

7 Is this a property of primes, or of density?

A natural objection to Conjecture 2 is that the chain grows dense enough for representability to be forced by counting alone. By the time we reach $\mathcal{Q}_k$ the pool $\mathcal{P}_k$ holds $4(k-1)$ members, so the number of sums of at most eight distinct members is $\binom{4(k-1)}{8}$, vastly exceeding the range of D . On that reading the conjecture would say nothing about primes. This section separates what the objection explains from what it does not.

What is structural

Two features of the data are not evidence about primes, and we do not claim them as such.

The first is the parity of the term counts. Every representation observed uses an *even* number of pool members — the counts are 2, 4, 6, 8 and never odd — and this is forced. Every member of every $\mathcal{Q}_j$ is an odd prime, so $D = t - m_k$ is a difference of two odd numbers and hence even; a sum of j odd numbers is congruent to j modulo 2; therefore j is even (Lemma 1(v)). No arithmetic beyond parity is involved.

The second is the bound $K = 8$ itself. To test it we ran the same solver on control chains built to have the same spacing as the real one, differing only in that their members are not prime:

- **jitter** — each quadruplet displaced inside its own local gap, preserving order and local density;
- **uniform** — each first member redrawn uniformly in the interval between its neighbours;
- **shift** — each first member displaced by a multiple of 30, preserving the residue class $11 \bmod 30$, and with it the parity argument above, while destroying primality.

Every control keeps the shape $(q, q + 2, q + 6, q + 8)$, the same counting function and the same base rule $m_k = q_{k-1}^{(9)}$, so any difference is attributable to the arithmetic of the members and not to how many there are. The **shift** control attains the same bound of eight terms and the same all-even distribution. The bound is a consequence of the residue class, not of primality.

What is not

chain	worst gap	exceptional gaps	occurrences	largest
real prime quadruplets	8	4	98	382
control, jitter (seed 1)	12	390	590	1700
control, jitter (seed 2)	11	103	141	1186
control, uniform (seed 1)	9	246	4455	2416
control, uniform (seed 2)	11	65	4221	1328
control, shift	8	26	733	3052

Table 3: All 28,387 quadruplets of each chain below 10^9 , 28,386 first-member gaps, under the base rule of Section 2. “Exceptional gaps” counts distinct values of D admitting no purely additive representation; “occurrences” counts the first members that meet one.

The exceptions are a different matter. The real chain has four, the best control has 26 — more than six times as many — and every control admits exceptions several times larger than the real chain’s maximum of 382. The contrast survives every control family we tried, including the one that reproduces the bound and the parity exactly.

Density and residue structure together predict that exceptions should be *rare*. They do not predict that there should be only four, that all four should be small, or that the set should close: no new exceptional gap appears after $\mathcal{Q}_{435}$, while 28,000 further quadruplets add none. That is the part of Conjecture 2 we regard as genuinely arithmetic, and the part we would most like to see explained.

8 What the conjecture says about the next quadruplet

It is tempting to read Conjecture 2 as a recipe: take the largest member of the last known quadruplet, add a short sum of earlier members, and the next quadruplet appears. We want to be precise about what is and is not true here.

What it gives. Every quadruplet after the first carries a *certificate* over its predecessors: a short equation whose terms are earlier chain members, with the base term identifying the previous quadruplet. The chain is therefore self-referential in a strong sense: nothing outside $\mathcal{Q}_0, \dots, \mathcal{Q}_{k-1}$ is needed to write down $\mathcal{Q}_k$ once it is known. The certificate also turns the search for the next quadruplet into a bounded additive search: if the conjecture holds with K terms, then the first member of $\mathcal{Q}_k$ is of the form $m_k + D$ with $D \equiv 22, 82, 112, 172$ or $202 \pmod{210}$, by Lemma 1(iv), and D a sum of at most K distinct pool members, and the candidate can be tested by four primality tests. The utility does exactly this in reverse, producing the certificate after the sieve has found the quadruplet.

What it does not give. The candidates produced by the additive search are far more numerous than quadruplets: near 10^9 the number of two-term sums of pool members below a typical gap is already in the thousands, and almost every admissible D has some short representation. So the conjecture does not point at the next quadruplet, it only says where it will be found once it exists, and in practice a sieve remains the faster way to find it. Nor does the conjecture imply that a next quadruplet exists; infinitude of prime quadruplets is a separate, open problem.

Why representations are plentiful. The Bateman–Horn heuristic predicts roughly $C_4 y / (\log y)^4$ quadruplets below y , and so four times as many pool members. The number of j -element subsets of pool members below D grows like

$$\frac{1}{j!} \left(\frac{4C_4 D}{(\log D)^4} \right)^j,$$

while their sums range over $O(D)$ even numbers, so for $j \geq 2$ the expected number of j -term representations of a given gap grows without bound as D grows, and grows faster the larger j is. This is the reason the mean minimal length stays flat, and it suggests the answer to the following question is yes.

Question 1. *Is every sufficiently large gap a sum of two distinct earlier members? Equivalently, do the lengths 4, 6, 8 in Table 1 eventually disappear? Lemma 1 leaves room for a yes: of the twenty-eight even classes modulo 210 lying over the four gap classes modulo 30, eight contain no sum of two members at all. A two-term sum is $(p + \delta) + (p' + \delta')$ with $p, p' \equiv 2, 3, 4 \pmod{7}$ by Lemma 1(ii), so $p + p'$ reaches only $\equiv 4, 5, 6, 0, 1 \pmod{7}$, and $\delta + \delta'$ is fixed by the class modulo 30; two classes modulo 7 are missed in each case. The twenty classes the lemma allows for gaps are exactly the twenty that survive.*

Question 2. *Is a royal value ever needed again beyond the four exceptional gaps, i.e. is Conjecture 2 true with exactly these exceptions and $K = 8$, or even $K = 6$ beyond some point?*

Question 3. *The same construction applies to any prime constellation, for instance twin primes or sexy prime triplets, with the corresponding admissible residues. Which constellations admit a bounded chain, and with what K ?*

Question 4. *The exceptional gaps are exactly the coverage holes of $\mathcal{Q}_1 \cup \mathcal{Q}_2$. Is there a simple description of the set of gaps that are not sums of pool members at the moment they are first met, for the general constellation?*

9 Availability and reproducibility

The utility, the checker and the statistics script are released in the folder LC of the repository

<https://github.com/SPAlgorithm/LE>

The commands below rebuild everything reported here from scratch.

```
python3 lc.py --upto 999999999 --cache paper/qarray_1e9.json # 43 s
python3 paper/check.py paper/qarray_1e9.json # 0 bad
python3 paper/stats.py paper/qarray_1e9.json # all figures
python3 lc.py 25 -AME --cache paper/qarray_1e9.json # the examples
python3 paper/residues.py paper/quads_1e9.csv # Lemma 1, 4 s
python3 paper/density.py 28387 # Section 7, ~5 min
```

The dataset `qarray_1e9.json` (23,085,594 bytes) has SHA-256

2d574932d1cbf0ff2692bfc8b2eab065d2ae35079255598bd0b3c89ecc0e5ee

and `quads_1e9.csv` has one row per member (113,548 rows, Q_1 included): for a first member the base, the gap, the four A/M/E1/E2 equations (the base-free M1/E3 equations are in the JSON only) and the three minimal lengths; for the other three members of a quadruplet only their fixed form. The control chains of Section 7 are generated inside `density.py` from a fixed seed, so its table is reproducible without any further data. Interactive use: `python3 lc.py` asks for a number of quadruplets; `-M`, `-E`, `-AME` select the ways; `--all` lists all four members in each heading; `-v` shows which quadruplet each term comes from; `-d N` prints a derivation of any integer N as in Section 4, extending the chain first if N lies beyond the cache.

Videos. Two narrated explanations accompany this note: *Ladhe’s Quad Conjecture: A New Pattern in Prime Quadruplets* (28 min, https://youtu.be/_XS0JOYj77Q), which covers Sections 3–8, and *Build Any Number From Prime Quadruplets* (11 min, <https://youtu.be/JNWvt3hS540>), which walks through the coverage proposition of Section 4 by hand.

App. The authors’ free iOS app *LE-Games* contains an interactive companion to this note: the chain itself with every way of writing each first member, a matching game (*QuadQuest*) whose every board is an equation of the chain, and a printable practice sheet: <https://apps.apple.com/app/id6767880385>.

Conflict of interest. The authors declare no competing interests related to this note.

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A The first 25 quadruplets

For each member: the quad index k , the member t , the base m_k , the gap D , the additive equation stored by the utility (base first), its length including the base, the minimal pure length K of the gap (“–” for the exceptional gaps), and, when the gap was met before, the quad where its equation was first found. Q_1 is written from royal members only; the second, third and fourth member of every later quadruplet appear in their fixed form $p+2$, $p+(2 \times 3)$, $(p+6)+2$, which has no K and no source.

k	t	m_k	D	additive equation	len	K	from
1	11			$(2*3) + 5$	2		
1	13			$(2*5) + 3$	2		
1	17			$7 + 5 + 3 + 2$	4		
1	19			$(2*7) + 5$	2		
2	101	19	82	$19 + 17 + 13 + 11 + (5*7) + (2*3)$	6	–	
2	103	101	2	$101 + 2$	2		
2	107	101	6	$101 + (2*3)$	2		
2	109	107	2	$107 + 2$	2		
3	191	109	82	$109 + 19 + 17 + 13 + 11 + (3*5) + 7$	7	–	2
3	193	191	2	$191 + 2$	2		
3	197	191	6	$191 + (2*3)$	2		
3	199	197	2	$197 + 2$	2		
4	821	199	622	$199 + 197 + 193 + 191 + 19 + 17 + 5$	7	6	
4	823	821	2	$821 + 2$	2		
4	827	821	6	$821 + (2*3)$	2		
4	829	827	2	$827 + 2$	2		
5	1481	829	652	$829 + 199 + 197 + 193 + 19 + 17 + 13 + 11 + 3$	9	8	
5	1483	1481	2	$1481 + 2$	2		
5	1487	1481	6	$1481 + (2*3)$	2		
5	1489	1487	2	$1487 + 2$	2		
6	1871	1489	382	$1489 + 199 + 109 + 19 + 17 + 13 + 11 + 7 + 5 + 2$	10	–	
6	1873	1871	2	$1871 + 2$	2		
6	1877	1871	6	$1871 + (2*3)$	2		
6	1879	1877	2	$1877 + 2$	2		
7	2081	1879	202	$1879 + 199 + 3$	3	2	
7	2083	2081	2	$2081 + 2$	2		
7	2087	2081	6	$2081 + (2*3)$	2		
7	2089	2087	2	$2087 + 2$	2		
8	3251	2089	1162	$2089 + 829 + 199 + 109 + 19 + (2*3)$	6	6	
8	3253	3251	2	$3251 + 2$	2		
8	3257	3251	6	$3251 + (2*3)$	2		
8	3259	3257	2	$3257 + 2$	2		
9	3461	3259	202	$3259 + 199 + 3$	3	2	7
9	3463	3461	2	$3461 + 2$	2		
9	3467	3461	6	$3461 + (2*3)$	2		
9	3469	3467	2	$3467 + 2$	2		
10	5651	3469	2182	$3469 + 2087 + 19 + 17 + 13 + 11 + (5*7)$	7	2	
10	5653	5651	2	$5651 + 2$	2		
10	5657	5651	6	$5651 + (2*3)$	2		
10	5659	5657	2	$5657 + 2$	2		
11	9431	5659	3772	$5659 + 3469 + 199 + 101 + 3$	5	4	
11	9433	9431	2	$9431 + 2$	2		
11	9437	9431	6	$9431 + (2*3)$	2		
11	9439	9437	2	$9437 + 2$	2		
12	13001	9439	3562	$9439 + 3467 + 19 + 17 + 13 + 11 + (5*7)$	7	2	
12	13003	13001	2	$13001 + 2$	2		
12	13007	13001	6	$13001 + (2*3)$	2		

k	t	m_k	D	additive equation	len	K	from
12	13009	13007	2	13007 + 2	2		
13	15641	13009	2632	13009 + 2089 + 199 + 197 + 109 + 19 + 17 + 2	8	6	
13	15643	15641	2	15641 + 2	2		
13	15647	15641	6	15641 + (2*3)	2		
13	15649	15647	2	15647 + 2	2		
14	15731	15649	82	15649 + 19 + 17 + 13 + 11 + (3*5) + 7	7	–	2
14	15733	15731	2	15731 + 2	2		
14	15737	15731	6	15731 + (2*3)	2		
14	15739	15737	2	15737 + 2	2		
15	16061	15739	322	15739 + 199 + 109 + 11 + 3	5	4	
15	16063	16061	2	16061 + 2	2		
15	16067	16061	6	16061 + (2*3)	2		
15	16069	16067	2	16067 + 2	2		
16	18041	16069	1972	16069 + 1877 + 19 + 17 + 13 + 11 + (5*7)	7	2	
16	18043	18041	2	18041 + 2	2		
16	18047	18041	6	18041 + (2*3)	2		
16	18049	18047	2	18047 + 2	2		
17	18911	18049	862	18049 + 829 + 19 + 11 + 3	5	4	
17	18913	18911	2	18911 + 2	2		
17	18917	18911	6	18911 + (2*3)	2		
17	18919	18917	2	18917 + 2	2		
18	19421	18919	502	18919 + 199 + 197 + 103 + 3	5	4	
18	19423	19421	2	19421 + 2	2		
18	19427	19421	6	19421 + (2*3)	2		
18	19429	19427	2	19427 + 2	2		
19	21011	19429	1582	19429 + 1487 + 19 + 17 + 13 + 11 + (5*7)	7	2	
19	21013	21011	2	21011 + 2	2		
19	21017	21011	6	21011 + (2*3)	2		
19	21019	21017	2	21017 + 2	2		
20	22271	21019	1252	21019 + 829 + 199 + 197 + 19 + 5 + 3	7	6	
20	22273	22271	2	22271 + 2	2		
20	22277	22271	6	22271 + (2*3)	2		
20	22279	22277	2	22277 + 2	2		
21	25301	22279	3022	22279 + 2089 + 829 + 101 + 3	5	4	
21	25303	25301	2	25301 + 2	2		
21	25307	25301	6	25301 + (2*3)	2		
21	25309	25307	2	25307 + 2	2		
22	31721	25309	6412	25309 + 5659 + 199 + 197 + 193 + 109 + 19 + 17 + 13 + (2*3)	10	4	
22	31723	31721	2	31721 + 2	2		
22	31727	31721	6	31721 + (2*3)	2		
22	31729	31727	2	31727 + 2	2		
23	34841	31729	3112	31729 + 2089 + 829 + 191 + 3	5	4	
23	34843	34841	2	34841 + 2	2		
23	34847	34841	6	34841 + (2*3)	2		
23	34849	34847	2	34847 + 2	2		
24	43781	34849	8932	34849 + 5659 + 3259 + 11 + 3	5	4	
24	43783	43781	2	43781 + 2	2		
24	43787	43781	6	43781 + (2*3)	2		
24	43789	43787	2	43787 + 2	2		
25	51341	43789	7552	43789 + 5659 + 1879 + 11 + 3	5	4	
25	51343	51341	2	51341 + 2	2		
25	51347	51341	6	51341 + (2*3)	2		
25	51349	51347	2	51347 + 2	2		

B The four exceptional gaps

Gaps that are not a sum of distinct pool members at the moment they are first met, with the additive (A), multiplicative (M) and exponential (E1) representations of the gap stored by the

utility.

D	k	t	A	M	E1
22	169	1006331	$19 + 3$	$(11*2)$	$(3\sim 2) + 13$
82	2	101	$17 + 13 + 11 + (5*7) + (2*3)$	$(17*3) + (13*2) + 5$	$(7\sim 2) + (11*3)$
172	435	3512231	$109 + 19 + 17 + 13 + 11 + 3$	$(19*7) + (17*2) + 5$	$(13\sim 2) + 3$
382	6	1871	$199 + 109 + 19 + 17 + 13 + 11 + 7$ $+ 5 + 2$	$(191*2)$	$(19\sim 2) + (7*3)$

C Notation

$\mathcal{Q}_0$	the royal quad $(2, 3, 5, 7)$
$\mathcal{Q}_k, k \geq 1$	the k -th prime quadruplet $(p, p + 2, p + 6, p + 8), p \geq 11$
$q_k^{(d)}$	the member of $\mathcal{Q}_k$ ending in digit $d \in \{1, 3, 7, 9\}$
$m_k = q_{k-1}^{(9)}$	the base of $\mathcal{Q}_k$
$D = p_k - m_k$	the gap of the first member of $\mathcal{Q}_k$
$\mathcal{P}_k = \mathcal{Q}_1 \cup \dots \cup \mathcal{Q}_{k-1}$	the pool available to $\mathcal{Q}_k$
$\mathcal{R}, \mathcal{R}_+$	royal values: all (30, every one flat), additive (11)
$\mathcal{R}^\pm$	non-negative royal values with $+, -, \times$ (83)
K	minimal length of a representation of a gap
C_k, T_k	integers with a derivation over $\mathcal{Q}_0.. \mathcal{Q}_k$; largest T with $[0, T] \subseteq C_k$
S_k	$S_3 = T_3, S_k = S_{k-1} + 4p_k + 16$: the proven lower bound for T_k